

Financial Crime Operations Analysis

Context

This report was developed as part of a performance and risk analysis simulation. It demonstrates the use of data analytics (SQL and Excel) to resolve operational bottlenecks and apply forensic analysis techniques to identify regulatory compliance gaps.

Project Overview

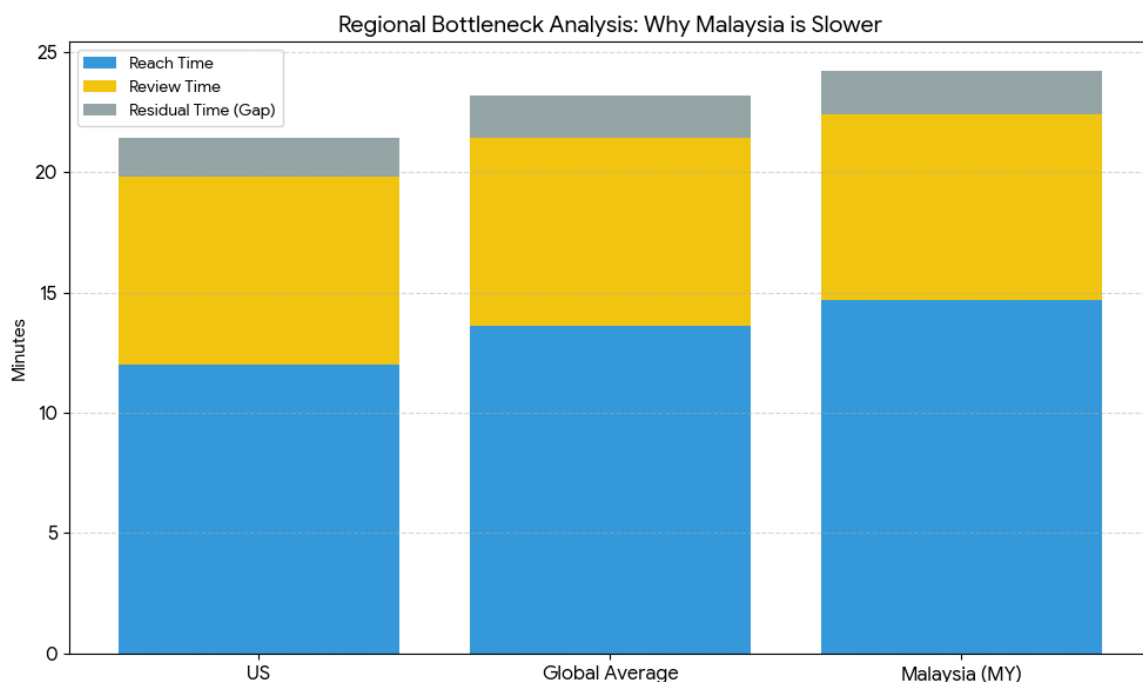
This project focuses on analysis of a global financial crime screening operation. The objective was to analyze large-scale performance data to identify operational bottlenecks and conduct a deep-dive quality audit of individual case logs to detect regulatory risks and system vulnerabilities.

Quantitative Performance Analysis

Using a dataset of 60 global screening cases, I performed a regional KPI benchmark. By analyzing the "Average Resolution Time," I identified significant variances in operational efficiency.

Key Findings:

- **Regional Bottleneck:** The Malaysia hub exhibited an average resolution time of 30.5 minutes, which is 41% slower than the US hub (21.5 minutes) and 20% slower than Estonia (25.5 minutes).
- **Operational Insight:** This variance indicates that while global processes are standardized, the Malaysia region requires targeted training or a review of local system latency to meet global "Reach Speed" targets.



Analytical Note:

The visualization above highlights a significant performance variance between global hubs. While the US and Estonia regions operate within a high-efficiency window (21.5–25.5 minutes), the Malaysia hub acts as a primary operational bottleneck with a 41% increase in resolution time. This data suggests that the regional delay is likely due to either localized system latency or a need for specialized training on high-complexity regional match lists

Technical Methodology: SQL Data Extraction:

To ensure data integrity and replicable results, the following SQL logic was used to aggregate regional performance metrics from the raw screening dataset:

```
SELECT
    Location,
    COUNT(Case_ID) AS Total_Cases,
    ROUND(AVG(Resolution_Minutes), 2) AS Avg_Resolution_Time,
    ROUND(AVG(Reach_Minutes), 2) AS Avg_Reach_Speed
FROM Compliance_Global_Logs
WHERE Case_Date BETWEEN '2026-01-01' AND '2026-01-31'
GROUP BY Location
ORDER BY Avg_Resolution_Time DESC;
```

Qualitative Case Log Analysis

Beyond high-level data, I conducted a forensic analysis of individual case logs to ensure procedural integrity. In Case #123456789, I identified a critical "Asset Flight" risk caused by a sequence error in the trail.

The Evidence:

- **12:59:54 UTC:** Payment status transitioned to RELEASED.
- **13:02:02 UTC:** Case status transitioned to SOLVED.
- **13:19:22 UTC:** Match officially marked as FALSE POSITIVE.

The Risk Assessment: The evaluation reveals that the payment was released 3 minutes before the investigation was solvedand nearly 20 minutes before the sanctions match was officially cleared as a False Positive. This represents a severe regulatory vulnerability; if the match had been a "True Positive," the funds would have already left the institution before the risk was confirmed.

Event	Actual Time	Status	Compliance Requirement
Payment Released	12:59:54 UTC	 Early	Must be AFTER resolution
Case Solved	13:02:02 UTC	 OK	Must be BEFORE release
Match Cleared	13:19:22 UTC	 Late	Must be BEFORE release

Proposed System Control & Risk Mitigation

To prevent future regulatory breaches and eliminate human error, I propose the following technical solution:

- **Solution:** "Hard Stop" Logic: Implement a system-enforced dependency where the "Release Payment" function remains disabled (greyed out) in the investigator's UI until the "Sanctions Match" status is transitioned to "False Positive."
- **Objective:** To ensure 100% synchronization between case resolution and asset release, removing the risk of "Asset Flight" during the investigation window.
- **Business Impact:** Implementing this 'Hard Stop' logic reduces the probability of human-error-driven 'Asset Flight' to 0%, ensuring the firm remains in 100% alignment with global anti-money laundering (AML) and sanctions regulations.

Professional Skillset Demonstrated

- **Data Analytics:** SQL-style filtering and KPI benchmarking using regional datasets.
- **Technical Product Thinking:** Designing automated system controls to mitigate human error in high-stakes environments.
- **Process Integrity Analysis:** Identifying timestamp anomalies and workflow deviations across structured operational records.